Alluvial soil is the largest and most important soil group in India. Covering about 15 lakh sq. km or about 45.6% of the total land area of the country, these soils contribute the largest share of our agricultural wealth. The widest occurrence of the Alluvial soils is in the Indo-Gangetic plain starting from Punjab in the West to West Bengal and Assam in the East. They are also found in deltas of Mahanadi, Godavari, Krishna and Cauvery, where they are called deltaic alluvium. Along the coast, they are known as coastal alluvium. Some alluvial soils are found in Narmada and Tapi (Tapti) valleys. Geologically, the alluvium is divided into newer or younger Khadar and older Bhangar soils. Geographically, Black soils are spread over 5.46 lakh sq. km while the red soils occupy a vast area of about 3.5 lakh sq. km.

6. Which one of the following is the most widespread category of soils in India?

(a) Alluvial soils (b) Black soils
(c) Red soils (d) Forest soils
(e) None of the these

Chhattisgarh P.C.S. (Pre) 2017

Ans. (a)

See the explanation of the above question.

7. Old Kachhari clay of Gangetic plain is called :

(a) Bhabar (b) Bhangar
(c) Khadar (d) Khondolyte

41st B.P.S.C. (Pre) 1996

Ans. (b)

Geologically, the alluvium of the great plain of India is divided into newer or younger khadar and older bhangar soils. Bhangar is found at a higher level beyond the reach of floods. Bhangar soil contains calcareous concretions and is generally pale reddish brown in colour.

8. Which of the following types of soil has minimum water retention capacity?

(a) Alluvial sand soil (b) Loamy sand soil
(c) Clayey loam soil (d) Loamy soil

Uttarakhand P.C.S. (Pre) 2003

Ans. (a)

Alluvial sand soil has the minimum water retention capacity because soil with limited water holding capacity (i.e., sandy loam) reaches the saturation point much sooner than a soil with a higher water holding capacity (i.e., clay loam).

9. Which soil particles are present in loamy soils?

(a) Sand particles
(b) Clay particles
(c) Silt particles
(d) All types of particles

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

Generally, Loam soil has 40% sand particles, 18% clay particles and 42% silt particles. Thus, option (d) is correct.

10. Which of the following soil has the lowest pH value?

(a) Bhatha soil
(b) Dorsa soil
(c) Matasi soil
(d) Kanhar soil

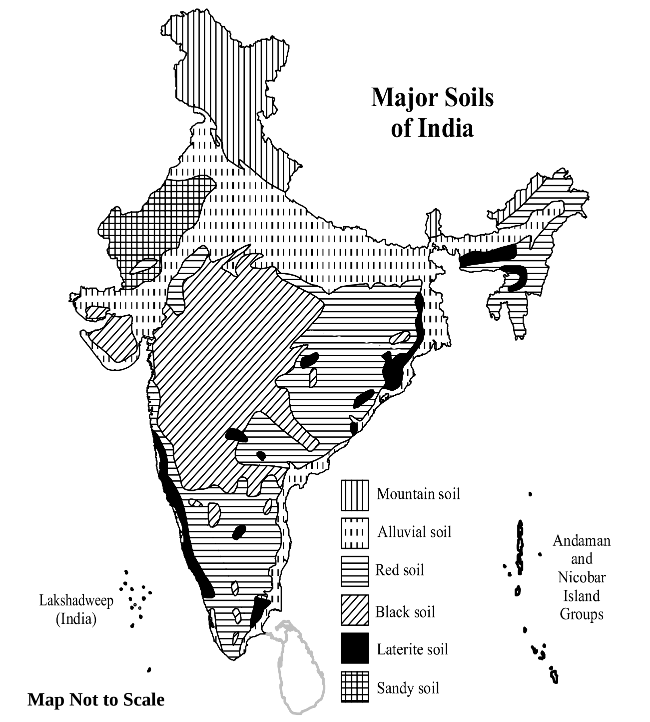
Chhattisgarh P.C.S. (Pre) 2024

Ans. (a)

Among the given options, the local names of Chhattisgarh's soils are mentioned. The pH values of these soils vary from one region to another based on their composition. Based on the general description of these soils, Bhatha soil (Laterite) has the lowest pH value, approximately 6.5, making it acidic in nature. This soil is leached and found in higher geographical regions. It ranges in texture from sandy loam to gravelly sand soil, and it is mostly red or red yellow in colour often classified as barren land.

iv. Soils : Miscellaneous

Red Soil - Red Soil in India, is formed by weathering of ancient crystalline igneous and metamorphic rocks. *The higher concentration of iron is responsible for the red colour of the soil. *These soils are found in regions of low rainfall and are obviously more leached than **Laterite soils.** The red soils cover a large portion of land in India (approximately 3.5 lakh sq. km). *Red soils are **poor in Phosphorus, Nitrogen and Lime content.** They are found in the Indian States of WesternTamil Nadu, Southern Karnataka, Southern Maharashtra, north-eastern Andhra Pradesh and some parts of M.P., Chhattisgarh, Odisha, Telangana and Chotanagpur (Jharkhand). The soil develops a reddish colour due to a wide diffusion of **iron** in crystalline and metamorphic rocks. It appears **yellow when it occurs in a hydrated form**.



Desert Soil - The Desert soil covers an area of approx. 1.42 lakh sq. km. The rainfall in these areas ranges from **less than 50 cm.** The **temperature is very high.** The avg. temperature of soil is 35°C. These soils are formed in dry climatic condition. They are **found in Rajasthan, Saurashtra (Gujarat), Kutch, Southern Punjab.** They contain a low amount of organic matter and nitrogen. *Legume crops enrich the soils with nitrogen content by atmospheric nitrogen fixation. Plants contributing to nitrogen fixation include **Soybeans**, **alfalfa, Lupins, Peanuts** and **Pulses**.

1. Match List-I with List -II and select the correct answer using the code given below :

List - I (Soil)	**List - II (State)**
A. Alluvial	**1. Rajasthan**
B. Black	**2. Uttar Pradesh**
C. Red	**3. Maharashtra**
D. Desert	**4. Andhra Pradesh**

Code :

	A	B	C	D
(a)	1	4	3	2
(b)	2	3	4	1
(c)	4	2	1	3
(d)	3	4	2	1

U.P.R.O./A.R.O. (Pre) 2021

Ans. (b)

The correct match is as follows :

(Soil)	(State)
Alluvial	Uttar Pradesh
Black	Maharashtra
Red	Andhra Pradesh
Desert	Rajasthan

2. Soils of western Rajasthan have a high content of:

(a) Aluminum (b) Calcium
(c) Nitrogen (d) Phosphorus

I.A.S. (Pre) 1993

Ans. (b)

The districts of Jaisalmer, Bikaner, Barmer, Jalore, Jodhpur, Jodhpur Gramin, Phalodi Sri Ganganagar, Sirohi, Jhunjhunu, Pali and Sikar etc. of western Rajasthan have alkaline and saline soils with a calcareous base. There is some nitrate concentration in the soil of these regions. Hence, soils of western Rajasthan have a high content of calcium.

3. Which one of the following crop enriches the Nitrogen content in soil ?

(a) Potato (b) Sorghum
(c) Sunflower (d) Pea

I.A.S. (Pre) 1994

Ans. (d)

Legume crops enrich the soil with nitrogen content through atmospheric nitrogen fixation. Plants that contribute to nitrogen fixation include clover, soybeans, alfalfa, lupins, peanuts, and rooibos, etc. Thus, option (d) is correct.

4. Which one of the following crops is grown to improve the soil fertility ?

(a) Wheat (b) Rice
(c) Black Gram (Urd) (d) Sugarcane

R.A.S./R.T.S. (Pre) 1996

Ans. (c)

The crop of Black Gram (Urd) is grown to improve soil fertility. Occasionally, it is also used as fodder.

5. When you travel in certain parts of India, you will notice red soil. What is the main reason for this colour?

(a) Abundance of magnesium
(b) Accumulated humus
(c) Presence of ferric oxides
(d) Abundance of phosphates

I.A.S. (Pre) 2010

Ans. (c)

Most of the red soils have come into existence due to weathering of ancient crystalline igneous and metamorphic rocks. The red soils are short of lime, phosphates, nitrogen and humus, but are fairly rich in potash. In their chemical composition, they are mainly siliceous and aluminous. The reason for the red colour of the soil is the presence of ferric oxide.

6. Given below the two statements, one is labelled as : Assertion (A) and the other as Reason (R).
Assertion (A) Red soil denotes the second largest soil group of India.
Reason (R) : Red soil surrounds the black soil in the-south, east and north directions.
Select the correct answer from the codes given below:
(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) (A) is false, but (R) is true.
(c) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(d) (A) is true but (R) is false.

UPPSC (Pre) 2024

Ans. (c)

Red soil is the second - largest soil group in India after alluvial soil. It covers approximately 10.5% of the country's total area. Its distribution is mainly in western Tamil nadu, Karnataka, Northeastern Andhra Pradesh, Southern Maharashtra, Rajasthan, Eastern Madhya Pradesh, Bihar, West Bengal, Chhattisgarh, Telangana, Odisha and the Chhotanagpur Plateau (Jharkhand). Black Soil is Primarily found in the Deccan Plateau. Therefore, it is evident that red soil surrounds black soil from the south, east and north directions.
Thus, both (A) and (R) are correct, but (R) does not provide the correct explanation of (A).

7. The micronutrient maximum deficient in Indian soils, is :
(a) Copper (b) Iron
(c) Manganese (d) Zinc

U.P.P.C.S. (Spl) (Mains) 2004

Ans. (d)

Zinc is the micronutrient that is most deficient in Indian soils among the given options.

8. Assertion (A) : The Himalayan soils are rich in humus.
Reason (R) : The Himalayas have the largest area under forest cover.
Code :
(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

U.P. Lower Sub. (Spl) (Pre) 2004

Ans. (d)

The Himalayan soils **lack humus** while the Himalayas have the largest area under forest cover. Thus, Assertion (A) is false but Reason (R) is true.

9. Karewas soils, which are useful for the cultivation of Zafran (a local variety of saffron), are found
(a) Kashmir Himalaya (b) Garhwal Himalaya
(c) Nepal Himalaya (d) Eastern Himalaya
(e) None of the above/More than one of the above

64th B.P.S.C. (Pre) 2018

Ans. (a)

Karewa soil is found in the Kashmir Valley and is used for growing local saffron called Zafran.

10. Soil water available to plants is maximum in :
(a) Clay soil (b) silty soil
(c) sandy soil (d) loamy soil

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (a)

Clay soil has the highest water retention capacity. Therefore soil water available to plants is maximum in it. Clay soil has less than 50% silt, 50% clay and some amount of sand. This soil slows air ventilation, due to which it sustains water.

11. Which one of the following particles has less than 0.002 mm diameter?
(a) Clay (b) Silt
(c) Fine sand (d) None of the above

U.P.R.O./A.R.O. (Pre) 2016

Ans. (a)

The diameter of clay is less than 0.002 mm. Silt has a diameter of 0.002 mm to 0.05 mm and fine sand's diameter is 0.05 mm to 2 mm.